

How to Build, Export & Update Docker Tar Bundles

Step-by-Step Developer Guide: Building Tar Bundles from Scratch & Incremental App Updates

BUILD & DEPLOYMENT ARCHITECTURE OVERVIEW

AI CyberAuditBox exports a single air-gapped Docker archive (.tar) containing all 5 microservices, database schemas, and AI model weights. Developers can rebuild the full bundle or update specific containers using single PowerShell commands.

Part 1: Building the Docker Tar Bundle from Scratch (Developer Guide)

Step 1: Verify System Prerequisites

Ensure Docker Desktop is running on Windows/Linux and git repository is checked out.

Step 2: Place AI Model Weights

Ensure LLM GGUF weights reside inside data/models/ (e.g. llama-3.2-3b-instruct.Q4_K_M.gguf).

Step 3: Run Full Bundle Build Script

Open PowerShell in project root and execute the single automated build command:
powershell -ExecutionPolicy Bypass -File .\update_bundle.ps1 -Version 1.4

Step 4: Understand Automatic Build Steps

- The script automatically:
1. Builds aicyberauditbox-app:1.4 image from Dockerfile.app.
 2. Updates version tags in docker-compose.yml and docker-compose.customer.yml.
 3. Packs all 5 services into a single ~6.8 GB tar file (aicyberauditbox_bundle_v1.4.tar).

Part 2: Fast Update Command (Updating ONLY App Code without re-downloading models)

When modifying Python backend logic or UI static files (app.js/index.html), you do NOT need to rebuild the heavy database or LLM containers. Use the single fast update command:

```
powershell -ExecutionPolicy Bypass -File .\update_bundle.ps1 -Version 1.4 -Services app
```

Part 3: Customer Air-Gapped Deployment Guide (4 Simple Steps)

Step 1: Copy Deployment Files

Copy aicyberauditbox_bundle_v1.4.tar and docker-compose.customer.yml into a target server directory.

Step 2: Load Docker Tar Bundle

Open terminal in that folder and run:
docker load -i aicyberauditbox_bundle_v1.4.tar

Step 3: Launch Microservices

Start all 5 microservices in background mode:
docker compose -f docker-compose.customer.yml up -d

Step 4: Monitor First Startup

Wait ~10-14 minutes while local LLM weights load into RAM. Check health status:
docker compose -f docker-compose.customer.yml ps
(Status will switch from 'unhealthy/starting' to 'healthy' once ready. This is normal behavior.)

Step 5: Access System Web UI

Open browser: http://localhost:8000/ - Runs 100% offline with zero internet access!

USEFUL CONTAINER MANAGEMENT COMMANDS

- Stop Services: docker compose -f docker-compose.customer.yml stop
- Restart Services: docker compose -f docker-compose.customer.yml restart
- View Live Logs: docker compose -f docker-compose.customer.yml logs -f app
- Update Single App Container: docker compose -f docker-compose.customer.yml up -d --no-deps app